Pneumatic Cylinders Manual

Kaveri Motion Components · Rev C, Mar 2025

Contents and symbols

This manual covers installation, operation and maintenance. Read the Safety Instructions before working on the product. Symbols: WARNING marks a hazard that can cause injury; NOTE marks information that prevents damage.

Technical data

Design standard ISO 15552. Bore sizes 32, 40, 50, 63, 80 and 100 mm. Operating pressure 1 to 10 bar (0.1 to 1.0 MPa). Proof pressure 15 bar is a factory test value only; never operate above 10 bar. Medium: compressed air filtered to 40 micron, lubricated or non-lubricated. Temperature range -20 to +80 deg C with standard NBR seals; -10 to +150 deg C with FKM seal option (order suffix -V). Standard strokes 25 to 1000 mm. Strokes above 1000 mm up to 2000 mm and bores above 100 mm are non-catalogue and are quoted by applications engineering. Adjustable pneumatic cushioning at both ends. Magnetic piston for position sensors is standard.

Ports and rod threads

Connection data by bore size.

bore 32 mm	air port G1/8, piston rod diameter 12 mm, rod thread M10x1.25
bore 40 mm	air port G1/4, piston rod diameter 16 mm, rod thread M12x1.25
bore 50 mm	air port G1/4, piston rod diameter 20 mm, rod thread M16x1.5
bore 63 mm	air port G3/8, piston rod diameter 20 mm, rod thread M16x1.5
bore 80 mm	air port G3/8, piston rod diameter 25 mm, rod thread M20x1.5
bore 100 mm	air port G1/2, piston rod diameter 25 mm, rod thread M20x1.5

Theoretical cylinder force at 6 bar

Theoretical force at 6 bar operating pressure.

bore 32 mm	extend 483 N, retract 415 N
bore 40 mm	extend 754 N, retract 633 N
bore 50 mm	extend 1,178 N, retract 990 N
bore 63 mm	extend 1,870 N, retract 1,682 N
bore 80 mm	extend 3,016 N, retract 2,721 N
bore 100 mm	extend 4,712 N, retract 4,418 N

Force at other pressures scales linearly: F(p) = F(6 bar) x p / 6. Effective force is lower because of seal friction

Ordering code

Ordering code PC15-[bore]-[stroke]-[options]. Example PC15-63-200 is 63 mm bore, 200 mm stroke, NBR seals, magnetic piston. Option -V: FKM (Viton) seals for high temperature. Option -T: through piston rod. Example PC15-50-100-V is 50 mm bore, 100 mm stroke with FKM seals.

Mounting accessories

Foot mountings, flanges, clevis and trunnion mountings to ISO 15552 are available. Refer to the KMC accessory catalogue for dimensions.

Installation and commissioning

Align the piston rod with the load; use a floating joint where misalignment is possible. Do not apply side loads to the piston rod. Commission at a reduced pressure of 2 bar, then raise to working pressure and set the cushioning screws. Depressurise the system before connecting or disconnecting fittings.

Maintenance, lubrication and seal kits

PC15 cylinders are pre-lubricated for life when used with non-lubricated air. If lubricated air has been used once, lubrication must be continued permanently with ISO VG 32 oil, because the factory grease is washed out. Inspect the rod seal every 6 months. Replace seals after approximately 5000 km of travel or when leakage is visible. Seal kits with NBR seals: PC15-SK-32, PC15-SK-40, PC15-SK-50, PC15-SK-63, PC15-SK-80, PC15-SK-100. Seal kits with FKM seals carry the suffix -V, for example PC15-SK-50-V and PC15-SK-63-V. Seal kits are wear parts.

Troubleshooting

Jerky motion: check air supply, flow control setting and lubrication. Cylinder does not reach end position: supply pressure too low or load exceeds available force. Air leaking at the piston rod: rod seal worn, fit the seal kit for the bore size. Hard impact at end of stroke: turn the cushioning screw clockwise to increase cushioning, or reduce speed.